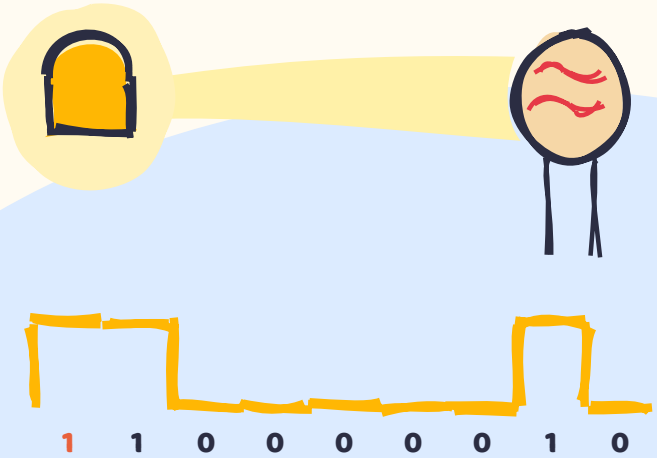
Young Innovators · Class 23

Li-Fi Messenger:
Sending Text with Light

No wires, no Wi-Fi. Just flashes.

ASCII | Light pulses | Decode



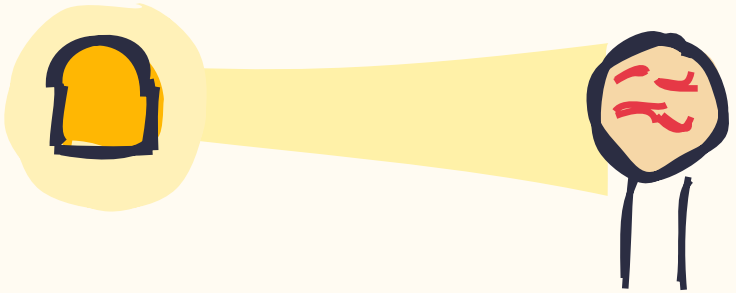
Did you know?
Li-Fi in labs has sent data at over 200 gigabits per second using LED light flickering too fast for eyes to see.

Let's think...

Could a lamp be your internet?

Light talks

Ships once flashed messages with lamps in Morse code. Your TV remote sends invisible infrared flashes. What if every ceiling light in school could send data, millions of flashes a second?



Impossible! **Too slow?** **Light = data!**

Today: you build a **Li-Fi messenger**: type a message on one Arduino and it flies across the desk as light to another.

Learning objectives

By the end of this class I can...

1	**Text → bits:** turn letters into **ASCII** numbers and **8-bit** binary bytes	**Proof:** I can write 'A' as 01000001
2	**Bits → light:** send bytes as **light pulses** using a start bit and a fixed **bit time**	**Proof:** my LED flashes a letter I typed
3	**Light → text:** receive with an **LDR**, decode the bits and measure **speed** and **errors**	**Proof:** my partner's Arduino printed my message

Objective 2 · Bits → light · video

Video time!



Forget Wi-Fi. Meet the new Li-Fi Internet

TED · about 7 minutes · youtu.be/iHWIZsIBj3Q

Watch for…

1. How can a **light bulb** send data?
2. Why is Li-Fi so **fast**?
3. Where could Li-Fi be **safer** than Wi-Fi?

Scan to watch

After the video: Name one room in our school where Li-Fi would be useful, and say why.

Objective 1 · Text → bits

Letters are numbers, numbers are bits

The idea

Computers store letters as numbers using **ASCII**: 'A' = 65, 'B' = 66, 'a' = 97, space = 32.

Each number fits in one **byte** = 8 bits:
65 = 64 + 1 → **0100 0001**

To send 'Hi' we send 2 bytes = 16 bits. Each bit is just **on** (1) or **off** (0).

Example: a secret code book

A → **01000001**

B → **01000010**

a → **01100001**

Two spies agree: 1 = apple, 2 = banana. Now they can send numbers instead of words. ASCII is the world's agreed code book for letters.

LEARN

Objective 2 · Bits → light

Timing is everything

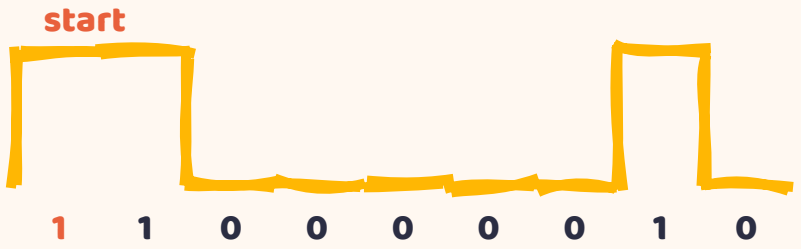
The idea

Light on = 1, off = 0. But how does the receiver know where one bit ends?

Agree a bit time: every bit lasts exactly **50 ms**.
Start bit: the line is normally dark; a flash ON says 'a byte is starting!'
Then 8 data bits, then **stop** (dark) for a rest.

Receiver waits 1.5 bit times, then samples in the **middle** of each bit.

Example: clapping to music



A band agrees on a tempo before playing. If the drummer speeds up and the guitarist doesn't, it's chaos. Sender and receiver must keep the same beat.

Objective 3 · Light → text

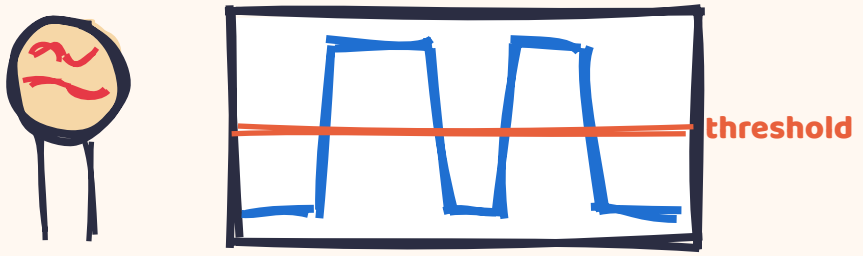
Receiving: threshold, speed, errors

The idea

An **LDR** changes resistance with light. In a divider on A0: bright → high reading, dark → low.
Threshold: halfway between your bright and dark readings.

Speed = bits per second = 1000 ÷ bit time (ms). 50 ms → 20 bps. LDRs are slow (tens of ms), so faster needs a **photodiode**. Real Li-Fi flickers millions of times a second.

Example: reading flashes in fog



In fog, a lighthouse flash is dimmer. Sailors still know it's 'on' if it's brighter than the darkness around it. That's a threshold.

Real world

Data on light



Li-Fi lamps
Offices and hospitals where radio is a problem



Fibre optic internet
Light in glass carries Pakistan's internet under the sea



TV remotes
Infrared flashes send button codes



Laser space links
Satellites talk to each other with lasers



Where would Li-Fi be better than Wi-Fi? Where would it be worse?

Light can't pass walls
That's a weakness (no signal next door) and a strength (hackers next door can't listen).

Quick quiz

Bits and light

Question	Answer
1. ASCII of 'C'?	
2. 67 in 8-bit binary?	
3. Bit time 50 ms. Bits per second?	
4. Why a start bit?	
5. Bright = 800, dark = 200. Threshold?	

Answers

Cover! 1: 67 · 2: 01000011 · 3: 20 · 4: tells the receiver a byte is beginning · 5: 500

Part 2 · Make it

Now let's put it to work!

You know the ideas. Time to play, build and test. Use these slides while you work:

1. Human Li-Fi
2. Our Li-Fi kit (per pair)
3. The sender
4. The receiver
5. Send messages, then push the speed
6. Speed vs errors
7. Network engineer challenges

MAKE

Class game

Human Li-Fi

How to play

1. Two teams across the room. Sender has a torch, receiver has paper.
2. Teacher claps a steady beat (the bit time). Torch on = 1, off = 0, one per clap.
3. Sender flashes a start flash, then 8 bits of a secret letter (use the ASCII card).
4. Receiver writes the bits and decodes. Then double the clap speed!

Then discuss

What went wrong when the beat got faster?



Big idea

Faster = more errors, unless the receiver is fast too.

MAKE

Build it

Our Li-Fi kit (per pair)

2 × Arduino Uno
One sender, one receiver

Bright white LED + 220 Ω
Or a laser module (teacher only)

LDR + 10 kΩ
Light sensor divider

Black straw / tube
Blocks room light

2 laptops
Serial Monitor on each

ASCII card
For checking by hand

Sender: LED + 220 Ω on pin 8. Receiver: 5V → LDR → A0 → 10 kΩ → GND.

Eyes
Never look into a laser module or shine it at anyone. The white LED is safe, but don't stare into it.

MAKE

Objective 2 · Bits → light · Build step 1 of 3

The sender

```
// SENDER
const int LED = 8;
const int T = 50;              // bit time in ms

void sendByte(byte b) {
  digitalWrite(LED, HIGH); delay(T);          // start bit
  for (int i = 0; i < 8; i++) {               // 8 data bits
    digitalWrite(LED, bitRead(b, i));         // lowest bit first
    delay(T);
  }
  digitalWrite(LED, LOW); delay(2 * T);       // stop / rest
}

void setup() {
  pinMode(LED, OUTPUT);
  Serial.begin(9600);
}

void loop() {
  if (Serial.available()) {
    char c = Serial.read();    // a letter you typed
    sendByte(c);
  }
}
```

bitRead

bitRead(b, i) gives bit i of the byte: 0 or 1.

Test

Type 'U' (01010101) and watch the LED blink in a steady pattern.

Idle = dark

Between letters the LED stays OFF, so a flash always means 'start'.

MAKE

Objective 3 · Light → text · Build step 2 of 3

The receiver

```
// RECEIVER
const int T = 50;
int TH = 500;                    // set from your own readings!

bool lit() { return analogRead(A0) > TH; }

void setup() { Serial.begin(9600); }

void loop() {
  if (lit()) {                   // start bit seen
    delay(T + T / 2);            // jump to the MIDDLE of bit 0
    byte b = 0;
    for (int i = 0; i < 8; i++) {
      if (lit()) bitSet(b, i);
      delay(T);
    }
    Serial.print((char)b);       // show the letter
    delay(T);                    // wait out the stop bit
  }
}
```

Threshold first

Print analogRead(A0) with the LED on, then off. Set TH halfway.

Middle

Sampling in the middle of each bit is safest: the edges are blurry.

Same T!

Both boards MUST use the same bit time.

MAKE

Objective 3 · Build step 3 of 3

Send messages, then push the speed

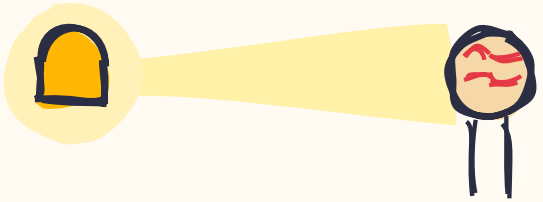
a Line up LED and LDR 5 to 20 cm apart. Put the LDR inside a black straw to block room light.

b Type 'HELLO' into the sender's Serial Monitor. Does it appear on the receiver?

c Try T = 50, 30, 20, 10 ms on **both** boards. Record speed and errors.

d Move them further apart. When does it stop working?

e Send a secret message to another pair across the table!



Check: Your partner reads your message on their screen.

Test like an engineer

Speed vs errors

Setting	Errors in 'HELLO WORLD'	Notes
T = 50 ms (20 bps)		
T = 30 ms (33 bps)		
T = 20 ms (50 bps)		
T = 10 ms (100 bps)		
Max working distance		cm

Debug checklist

Gibberish? Threshold wrong: print readings again.
Missing letters? Receiver too slow: increase T.
Works in the dark only? Shield the LDR with a tube.

Every real network trades speed against errors. Engineers add error checking (parity bits) to catch mistakes.

MAKE

Level up!

Network engineer challenges

Choose your challenge

- **Parity bit:** add a 9th bit so the receiver can detect a single flipped bit.
- **Faster:** swap the LDR for a phototransistor. How fast can you go?
- **Two-way:** add a second LED/LDR pair so both sides can chat.
- **Mirror:** bounce the beam off a mirror around a corner.

Think like an engineer

Why can't someone in the next room read your Li-Fi message? Is that always true?

Boss level

Send a small picture: 8 bytes of an 8×8 emoji, drawn on the receiver.

Objective check

Can I really do it? Prove it.

01	Write 'B' in ASCII and in binary.	○ ○ ○
02	Explain the start bit and the bit time. Why must both boards agree?	○ ○ ○
03	Show a message received. What speed worked best, and how many errors?	○ ○ ○

Rate yourself

Colour one circle each. Red: not yet. Amber: with help. Green: I can explain it to a friend.

Teacher: Li-Fi relay: pass a message from pair to pair across the room by light. Does it arrive unchanged?

Show what you know

What did we learn?

- [x] ASCII turns letters into numbers
- [x] 1 byte = 8 bits, sent one at a time
- [x] Start bit + fixed bit time keep sender and receiver in sync
- [x] Threshold decides on/off; sample in the middle
- [x] Faster = more errors unless the hardware is faster

Exit ticket: how many bits does it take to send the word 'Li-Fi'?

At home

Find 3 things at home that send data with light (remote, fibre router...).



Next class:
Python Code Breakers: Caesar to Vigenère!